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# On the application of 3-D scanning technology for the documentation and typology of lithic artifacts

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ABSTRACT

A 3-D optical scanner was used to obtain precise and complete representations of lithic artifacts. A computer algorithm, which was specially developed for the purpose, was used to position the artifacts in a way which enables the extraction of the standard metric parameters (length, width, width at 1/2 length, etc.). In this way, the ambiguities which affect the traditional manual measurements were eliminated. This new methodology creates accurate and objective databases. Several other parameters (center of mass position, volume, surface area) were also computed. The advantages of our method are illustrated by the analysis of 90 scanned Lower Paleolithic handaxes.

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## 1. Introduction

Lithic artifacts have fascinated scholars since the beginning of archaeological science (Evans, 1897; Leakey, 1951; Goodwin and van Riet Lowe, 1929), and various characteristics of these stone tools are considered essential tools for understanding and reconstructing our prehistoric past. Thus, the observed differences between lithic assemblages are often interpreted as indicators of diversity or evolutionary phases within prehistoric human cultures. Accordingly, a constructive comparison between lithic assemblages should be based on a multi-variant clustering, taking into account morphological and technological aspects. In this paper we concentrate on one of these features, namely, morphological *characterization*.

Since the early stages of prehistoric research, there were persistent attempts to develop a common set of measures – a language – in which the morphology of different lithic assemblages could be classified and compared. One can discern two broad approaches to these attempts: the *descriptive* and the *metric* approaches. The descriptive language attempts to capture the overall form of the object, by associating it with a familiar template. By using terms such as 'pear-shaped' or 'drop like', the descriptive language can characterize an assemblage according to the frequency of appearance of particular forms (e.g., Kleindienst, 1962; Tixier, 1956). The metric language attempts to *quantify* the morphology by a set of discrete metric measures, characterizing an assemblage by the distribution of these measures such as, e.g., the width-to-length ratio in the assemblage (e.g., Olszewski, 1986).

Both classification methods may be considered insufficient and subjective to a larger or lesser extent. The common reason for the failures of the two languages is that the spectrum of shapes of stone tools is so wide and complex that it can never be completely described in terms of several numbers or labels. The descriptive language is advantageous as it refers to the entire shape, or to global properties of the artifact, and not to local measures. Yet, its weakness stems from its being subjective, impressionistic, and cannot be quantified. Several attempts to overcome these drawbacks, by defining global attributes such as the *asymmetry* or *roughness*, were recently introduced (Saragusti, 2003; Saragusti et al., 2005).

The present study is mainly concerned with the metric approach. The definition and application of lithic metric measures appear to be rather problematic (Beck and Jones, 1989; Fish, 1978). This ambiguity is a source of frustration, expressed so eloquently by Gnaden and Holdaway (2000) in their discussion on variation in artifact recording:

> "There is a fundamental necessity to determine the level of variation attributable to the people who manufactured artifacts in the past as distinct from the variation introduced by modern-day archaeologist" (Gnaden and Holdaway, 2000, p. 745).

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A similar statement points to the source of difficulty:

> "An inspection of the digital image for this artifact, together with its other dimensions, reveals that there was probably a problem in the way the artifact was oriented for measurement… indicating that some of the observers were taking the wrong measurements" (Gnaden and Holdaway, 2000, p. 744).

We should emphasis that the source of ambiguity is not necessarily that "the observers were taking the *wrong* measurements". Rather, it may be due to an intrinsic difficulty, encountered in the study of uneven and asymmetric objects: there may be several different, but equally legitimate orientations of a lithic object. Each orientation produces a different set of metric measures, which clearly indicates that the shape of the object cannot be uniquely derived from the metric parameters. In particular, metric data retrieved by different scholars cannot offer a sound basis for a comparative study.

The main goal of the present work is to overcome the problems mentioned above and propose a new methodology for object orientation which is accurate, easy to implement and entirely *independent of the eye of the beholder*. The method is based on an objective algorithm that defines the positioning of an artifact, from which point on the unique metric measures follows without any ambiguity or uncertainty. In addition, we suggest that the positioning method chosen in the present work is preferred because it is practical and conceptually simple. Its ultimate acceptance test was conducted by applying it to a large variety of handaxes and other bifacial tools.

The method requires the use of a high precision 3-D digital camera, which records the complete artifact, and displays its digitized image as a triangulated point-cloud. This 3-D technology is used in various archaeological contexts. The most prominent use of this technology is in the study of pottery (Kampel et al., 2005; Mara and Sablatnig, 2005; Karasik and Smilansky, 2008; and references within), and in the study of architecture and landscape (Brown et al., 2001; Lambers et al., 2007). However, the application of the 3-D scanning for prehistoric lithic studies is scarce. Our algorithm uses the triangulated point-cloud image for finding the proper positioning of the artifact based on the artifact geometry. Then, the program computes all the metric parameters and the sections, traditionally used in lithic analyses. Finally, the traditional hand-drawing is substituted by the computerized result of the analysis. In addition, the digital image enables the computation of new and relevant parameters which are nearly impossible to be deduced using the traditional approach. In the present paper we shall discuss two examples, namely the volume and the position of the center of mass. Other parameters will be introduced and discussed in future publications.

A sample of 90 Acheulian handaxes is presented as a case study. These were randomly chosen from three Lower Palaeolithic sites in Israel: Umm Qatafa ($N = 26$), Ma'ayan Barukh ($N = 39$), and Nahal Zihor ($N = 25$). The handaxes were scanned and their digital images were analyzed using our novel method. The subset of handaxes from Ma'ayan Barukh is of particular methodological importance since the Ma'ayan Barukh assemblage was previously analyzed independently three times by the traditional manual method. We can thus compare the results obtained using the new method with the previously reported work.

The analysis of the assemblage will also exemplify the advantage of the use of 3-D scanning to extract measures that are only accessible with 3-D information. In particular, we shall focus on the volume, and the distance between the center of mass (CM) and the center of minimum enclosing cube (CMEC). It will be shown that these measures facilitate a sub-division of the assemblage into morphological groups which are well correlated with the clustering into temporal groups which were deduced independently.

## 2. Traditional documentation

The classification of bifaces is primarily based on a set of metric parameters which is expected to define their shape (morphology). The essential prerequisite for the analysis of bifaces is to position them in a fixed frame of reference. Traditionally, this is done in a way which can be schematically described as follows: first, one identifies an approximate symmetry plane which dissects the object and separates the faces of the artifact as shown in the left illustration of Fig. 1. The symmetry plane intersects the object at its "cutting edge" and the line of intersection is the main profile or boundary of the object. (The drawn profile is the vertical projection of the artifact on this plane). One then proceeds to identify the second symmetry plane which divides the object into (approximately) mirror symmetric parts and is perpendicular to the first plane. The right illustration of Fig. 1 demonstrates the vertical projection of the artifact on this plane. The intersection of the two planes is the vertical line seen in the left illustration of Fig. 1. Conventional analysis encases the object in the smallest rectangular frame (Roe, 1994), which can be visualized as an enclosing box with paces parallel to the symmetry planes, and whose dimensions can be defined as the length, maximal thickness and maximal width of the object, as shown in Fig. 1. Other metric measures such as the width at 1/5 of the height follow naturally. One could also section the object in any way to obtain other profiles as required. Technically, some observers extract measures while holding the handaxe, others construct a real auxiliary box, and position the handaxe inside it. It is important to stress again that the positioning of a *highly asymmetric* object as the common handaxe is not unique, and this is the core of the ambiguities and frustration described in Section 1. The intrinsic impossibility to find a "true" or "unique" positioning is a mathematical fact which cannot be circumvented. Drawings (mostly by hand) of selected handaxes from three views (two faces and side view) are used to provide further information about the artifacts surface. Occasionally, the drawings are replaced by 2D photographs from two or three directions (Fig. 2).

The parameters which are commonly recorded for each handaxe include (Fig. 1):

| | |
|---|---|
| Length ($L$) | Width at 4/5 length (B1) |
| Width ($W$) | Location of maximum thickness (L2) |
| Thickness ($T$) | Width at half length (W1) |
| Location of maximum width (L1) | Thickness at 1/5 length (T3) |
| Width at 1/5 length (B2) | Thickness at 4/5 length (T2) |
| Thickness at half length (T1) | |

They are measured in millimeters to the nearest mm except T1, which is measured to the nearest half mm (Roe, 1994). If the handaxe is asymmetrical, there will be a 'zone', rather than a single point, of maximum width. In this case the location of maximum width is measured from the butt's end to the midpoint of this zone (Roe, 1994).

A series of ratios is calculated from these metric attributes:

width/length ($W/L$);
thickness/width ($T/W$);
width at 4/5 length/width at 1/5 length (B1/B2);
location of maximum width/length (L1/$L$);
location of maximum thickness/length (L2/$L$).

Each of these ratios is considered as a relevant measure to a particular property of the handaxe. For example, the rationale behind the calculation of the thickness/width ratio is to provide a degree of handaxe modification. When thickness is plotted over width, the higher the ratio the more unrefined is the handaxe

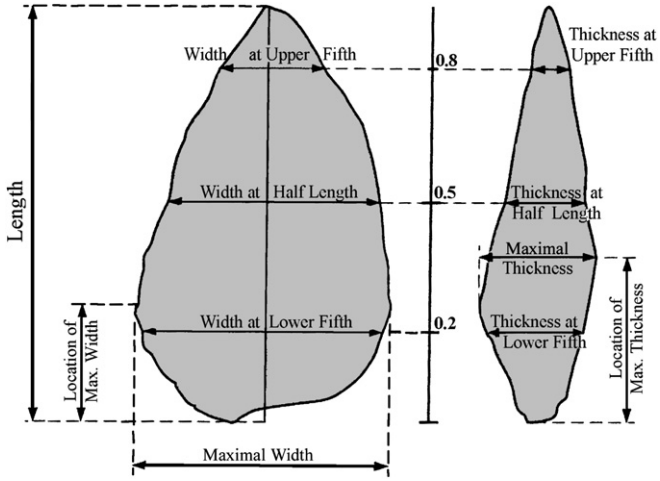


**Fig. 1.** Conventional positioning and basic measurement extraction (from Roe, 1964 and after Sharon, 2007).

analyzed. Bordes' (1961: pp.71–85) classification of handaxes is largely based on these ratios, which are also employed by Roe (1964, 1968) in his classification of British Lower Palaeolithic handaxes.

We have stated above that there is no unambiguous way to properly position an asymmetric and irregular object such as a handaxe. We shall now try to delve deeper into this problem. Fig. 3 illustrates a single handaxe in nine different orientations (Fig. 3a), obtained by turning the handaxe about the point (0,0) at 5° intervals in both directions. Three acceptable orientations with their resulting measures are shown in Fig. 3b. The variation in the width parameter with the tilt angle is shown in Fig. 3c and differences as high as ±18% of the mean width of the artifact are clearly seen. This ambiguity infiltrates into all the measures quoted for various assemblages and introduces variability and uncertainty which are entirely due to the subjective decision of the archaeologist who performs the measurements. Often different experienced archaeologists prefer different positioning.

Manual measurements include also human measuring errors. In order to obtain a quantitative estimate of the error accumulated as a result of the intrinsic and human factors, 20 handaxes from the site of Ma'ayan Barukh were measured by an experienced archaeologist (the 'observer'). The handaxes were held by hand and measured by a caliper. In the more complex measurements the observer was asked to repeat the manual measurements twice, and in Table 1 we summarize the mean deviations between his two trials. The deviations exceed the expected human error (Eerkens, 2000) and they are probably mostly due to the fact that a "proper" positioning does not exist.

We therefore conclude that the manual measurements and the comparative research which use them should be carefully scrutinized, and that possible intrinsic errors should be carefully eradicated.

## 3. Obtaining a 3-D scan

The lithic objects were scanned by the 3-D camera, manufactured by Polygon Technology (Darmstadt, Germany) (Fig. 4). Its operation is based on structured light projected on the artifact, recorded by two digital cameras. The scanning routine and the conversion of the data to a 3-D digital model are carried out by the program QTSculptor. Karasik and Smilansky (2008) recently presented a detailed account of this scanner and its application for the study of ceramics. Our scanning routine is quite similar but for a few features that are particular to lithic scanning. The lithic artifacts scanned in this study are made of flint which is harder to scan due to its surface reflectivity and its darker color. The complete surface of the artifact is obtained by scanning it in two settings. At each setting the artifact is clamped to the turntable (once at the distal and once at the proximal edge). The turntable exposes the

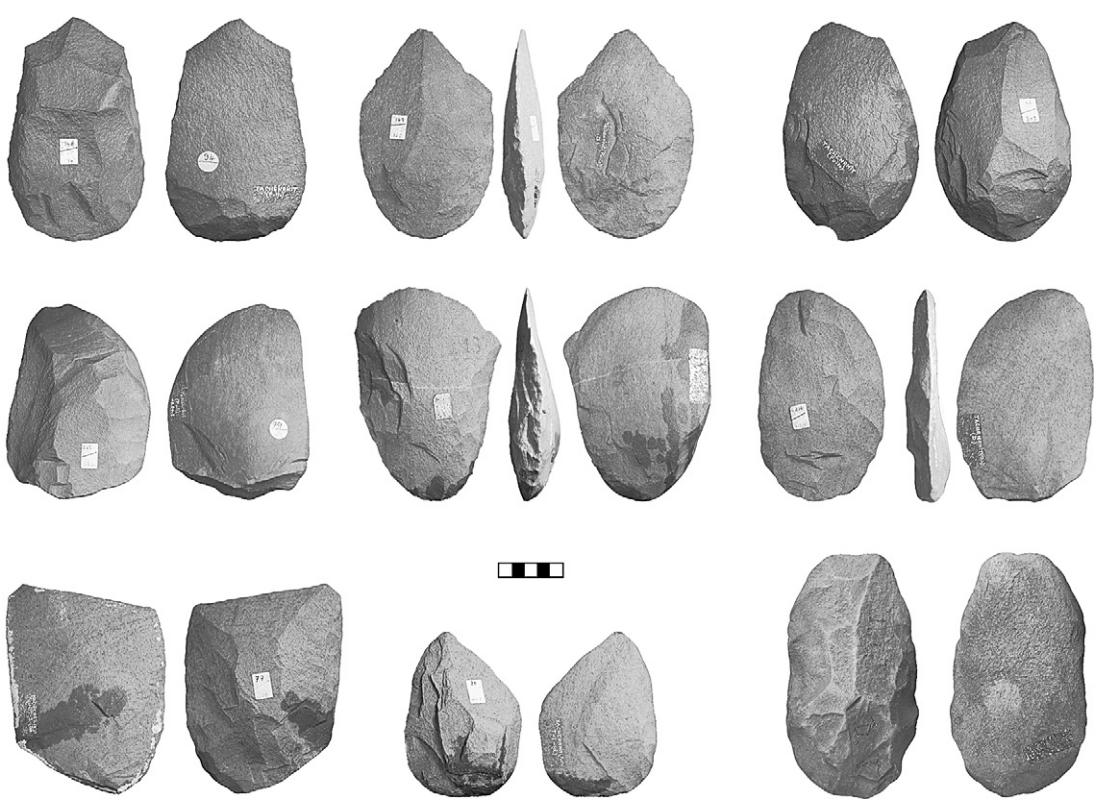

**Fig. 2.** Recently published 2D images of bifaces from Tachenghit (after Sharon, 2007).

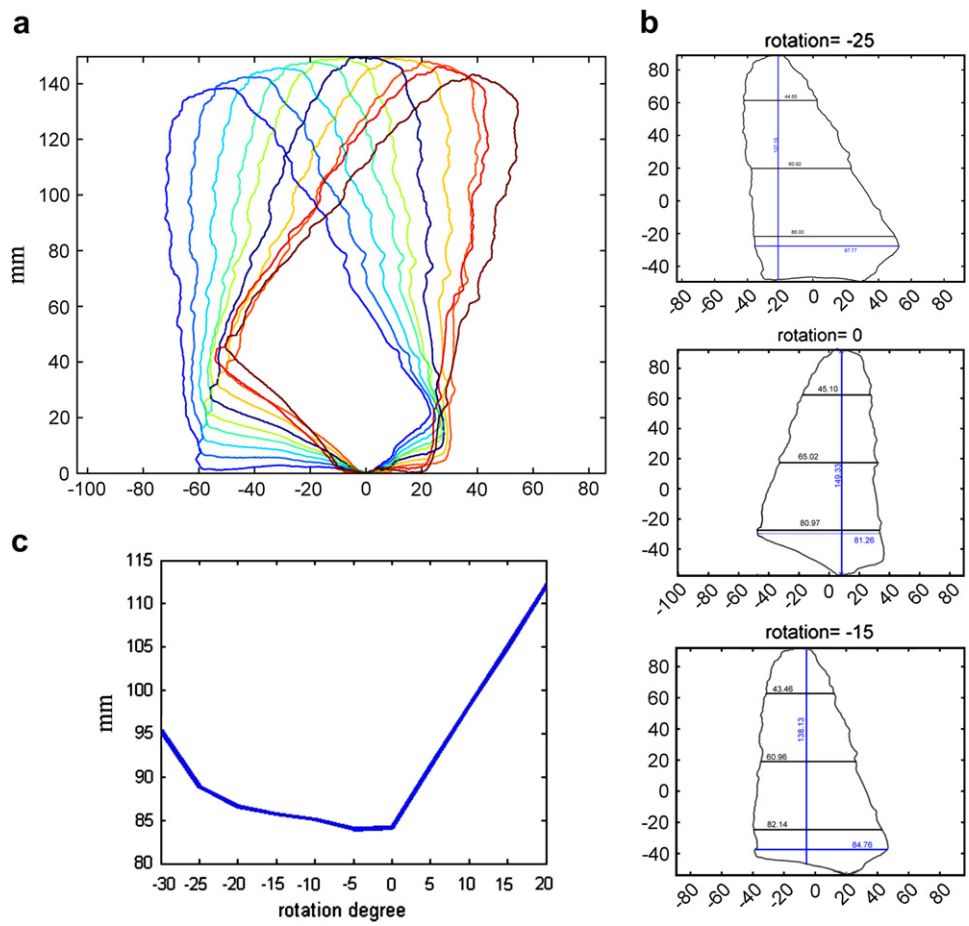


**Fig. 3.** (a) Various orientations of the handaxe. The angular step is 5°. (b) Three acceptable positioning and the extracted measures. (c) The variation of the width measure with the tilt angle.

object to the camera at 10 equally spaced (36°) intervals (compared to 6–8 times for ceramics). All the views in one setting form a "cluster". The two clusters are combined (registered) to form a closed triangular mesh. The entire surface is captured in both clusters, and this redundancy is used to obtain a higher precision of the 3-D representation. The number of triangles representing the digital image range from 20,000 to 500,000 depending on the nature of the surface and the required resolution, while the required memory is 2–50 MB, respectively.

An experienced operator produces complete digital 3-D images at the rate of 5–6 artifacts per hour (we employed students, not necessarily from Archaeology). Fig. 5 illustrates a two-dimensional view extracted from the digital record in two alternative displays. However, the scanned image is not yet positioned, and Section 4 will elaborate on the algorithm we developed to position the artifact and on extracting basic metric attributes.

**Table 1**
Mean percent error and standard deviation between two successive measurements of handaxe metric attributes

| Metric measurements | Mean percent error | Standard deviation |
|---|---|---|
| W1 – width at half length | 1.4 | 1.5 |
| T1 – thickness at half length | 2.8 | 3.7 |
| B1 – width at 4/5 length | 4.9 | 6.5 |
| T2 – thickness at 4/5 length | 7.4 | 6.6 |
| B2 – width at 1/5 length | 1.7 | 1.9 |
| T3 – thickness at 1/5 length | 4.0 | 3.1 |
| L1 – location of maximum width | 10.0 | 9.3 |
| L2 – location of maximum thickness | 6.4 | 5.9 |

## 4. How to position a scanned object?

We propose here a positioning procedure which depends only on the intrinsic properties of the object. Furthermore, it is mathematically rigorous and therefore provides an algorithm which is reproducible and independent of subjective judgment. Most important, the method was tested on a large variety of prehistoric flint tools, and it never failed to provide a satisfactory positioning as judged by experienced archaeologists.

To understand the idea, which is at the basis of the proposed positioning method, imagine that at each point on the surface of the object we attach a nail of unit length which is perpendicularly glued to the surface at the point of attachment. The resulting object looks like a hedgehog and it is illustrated in Fig. 6. It is quite clear that a large fraction of the nails is approximately perpendicular to the first symmetry plane which is the plane of the drawing in Fig. 1. (The perpendicular nails are seen as dots or very short intervals because they are "shortened" from the observer's direction of view).

The nails which lie in the plane of the picture are mostly concentrated about the active edge of the object. Focusing our attention to these nails only, we see that a larger fraction point in the up/down position, and a lesser fraction point in the left/right direction. Thus, by studying the distribution of nail directions on the surface of the object, we can determine symmetry planes which establish the object position. The advantage of this way of looking at the problem is that it can be quantified and turned into a computer algorithm. The mathematical formulation is given in Section 5.

The surface of the object is given as a triangulated point-cloud. Denote the number of triangles by $N$, and enumerate the triangles

**Fig. 4.** 3-D camera – Polygon Technology (Darmstadt, Germany).

by the index $i = 1, \ldots, N$. Denote the (outward pointing) normal to the $i$th triangle by $\vec{n}^{(i)} = (n_x^{(i)}, n_y^{(i)}, n_z^{(i)})$, and by $\sigma^{(i)}$ the area of this triangle. The surface area of the entire object is $A = \sum_{i=1}^{N} \sigma^{(i)}$. Our aim is to find a unit vector $\vec{h}$ which is parallel to most of the surface normals. That is, we would like to maximize

$$\chi(\vec{h}) \equiv \frac{1}{A} \sum_{i=1}^{N} \sigma^{(i)} \left( \vec{h} \cdot \vec{n}^{(i)} \right)^2$$

where $(\vec{h} \cdot \vec{n})$ stands for the scalar product. In other words, we search for the unit vector $\vec{h}$ that satisfies simultaneously the three equations:

$$\frac{\partial}{\partial h_x} \left[ \frac{1}{A} \sum_{i=1}^{N} \sigma^{(i)} \left( \vec{h} \cdot \vec{n}^{(i)} \right)^2 - \lambda (\vec{h} \cdot \vec{h}) \right] = 0,$$

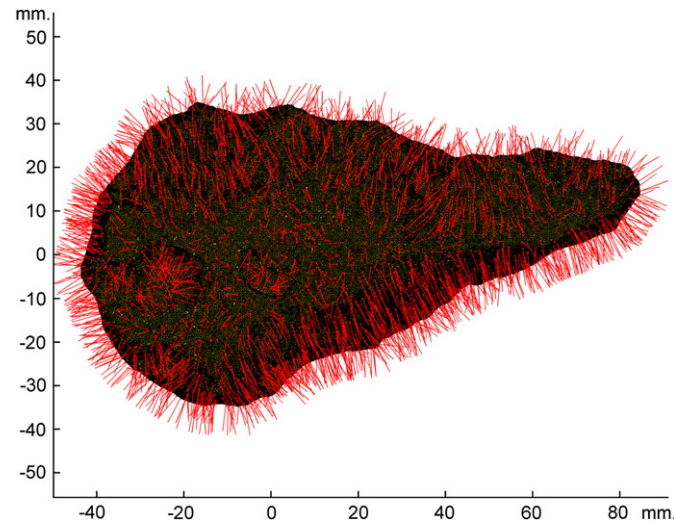


**Fig. 6.** The normal vectors to the complete surface of a handaxe.

$$\frac{\partial}{\partial h_y} \left[ \frac{1}{A} \sum_{i=1}^{N} \sigma^{(i)} \left( \vec{h} \cdot \vec{n}^{(i)} \right)^2 - \lambda (\vec{h} \cdot \vec{h}) \right] = 0,$$

$$\frac{\partial}{\partial h_z} \left[ \frac{1}{A} \sum_{i=1}^{N} \sigma^{(i)} \left( \vec{h} \cdot \vec{n}^{(i)} \right)^2 - \lambda (\vec{h} \cdot \vec{h}) \right] = 0.$$

The requirement is that $|\vec{h}| = 1$ is introduced by the Lagrange multiplier $\lambda$. Taking the derivatives and performing a few algebraic manipulations we find that $\vec{h}$ must satisfy the equation

$$T\vec{h} = \lambda \vec{h},$$

where $T$ is the *surface tensor*

$$T_{s,t} = \frac{1}{A} \sum_{i=1}^{N} \sigma^{(i)} n_s^{(i)} n_t^{(i)}; \quad s, t \in \{x, y, z\}.$$

This equation shows that $\vec{h}$ is an eigenvector of the surface tensor $T$, and $\lambda$ is the corresponding eigenvalue. It is not difficult to check that $T$ is symmetric and therefore its three eigenvalues are real and the corresponding eigenvectors are mutually orthogonal. The eigenvector with the largest eigenvalue maximizes $\chi(\vec{h})$ and therefore it provides the direction perpendicular to the major symmetry plane. The eigenvector corresponding to the second

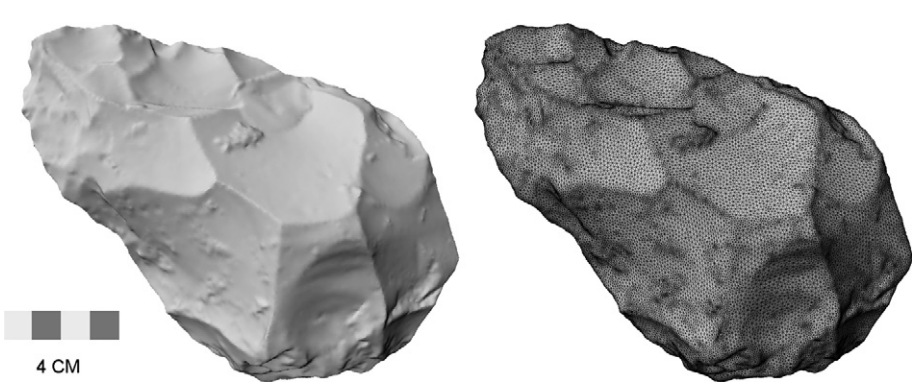


**Fig. 5.** Two images of a scanned handaxe. Shaded display (left) and point-cloud display (right).

eigenvalue is a local maximum, and it is perpendicular to the second symmetry plane.

The following comments might shed further light on the construction above:

- Consider, e.g., a spherical object. Then, the three eigenvalues are equal and there is no preferred direction for positioning the object, as expected.
- A surface of revolution (such as a cylinder) has two equal eigenvalues. The eigenvector corresponding to the single eigenvalue defines the axis of symmetry of the object.
- The continuum limit of the construction above consists of surface integrals which replace the corresponding sums.
- The fact that the normal vectors are of unit length, and because of the normalization which was chosen for the definition of *T*, it is easy to check that the sum of the eigenvalues is 1.

An alternative positioning method uses the eigenvectors of the *inertia tensor*, rather than the surface tensor defined above. The inertia tensor is a well known concept in classical mechanics, and it determines the way by which an asymmetric body responds to an applied torque (Goldstein, 1950). Assuming that the distribution of mass is uniform in lithic artifacts, the inertia tensor is defined by

$$S_{s,t} = \int_{\Omega} d^3\vec{r}(s \cdot t); \quad s,t \in \{x,y,z\}$$

Here, $\Omega$ is the volume of the artifact and $\vec{r} = (x,y,z)$ is the radius vector from the center of mass to the point in the artifact. The computation of the three-dimensional integral may be somewhat costlier than the computation of the normal tensor.

The eigenvectors of the inertia tensor, ordered by the magnitude of the corresponding eigenvalues, are used to position the object in the same way as described above for the surface tensor. As a matter of fact, the two sets of vectors are in most cases parallel and there is no *a priori* reason to prefer one over the other. In the present work we consistently used the eigenvectors of the inertia tensor for positioning.

After positioning, the handaxe is drawn and a section of the artifact perpendicular to the vertical axis can be added to the final object documentation shown in Fig. 7 (five views, scale and section). Additionally, basic measurements can be presented in a sketch form, which includes all the standard measurements traditionally presented in Lower Paleolithic handaxe studies (Fig. 8).

The 3-D information enables the extraction of important parameters which were never studied before. Prominent examples are the volume ($v$), the surface area ($A$), the coordinates of the

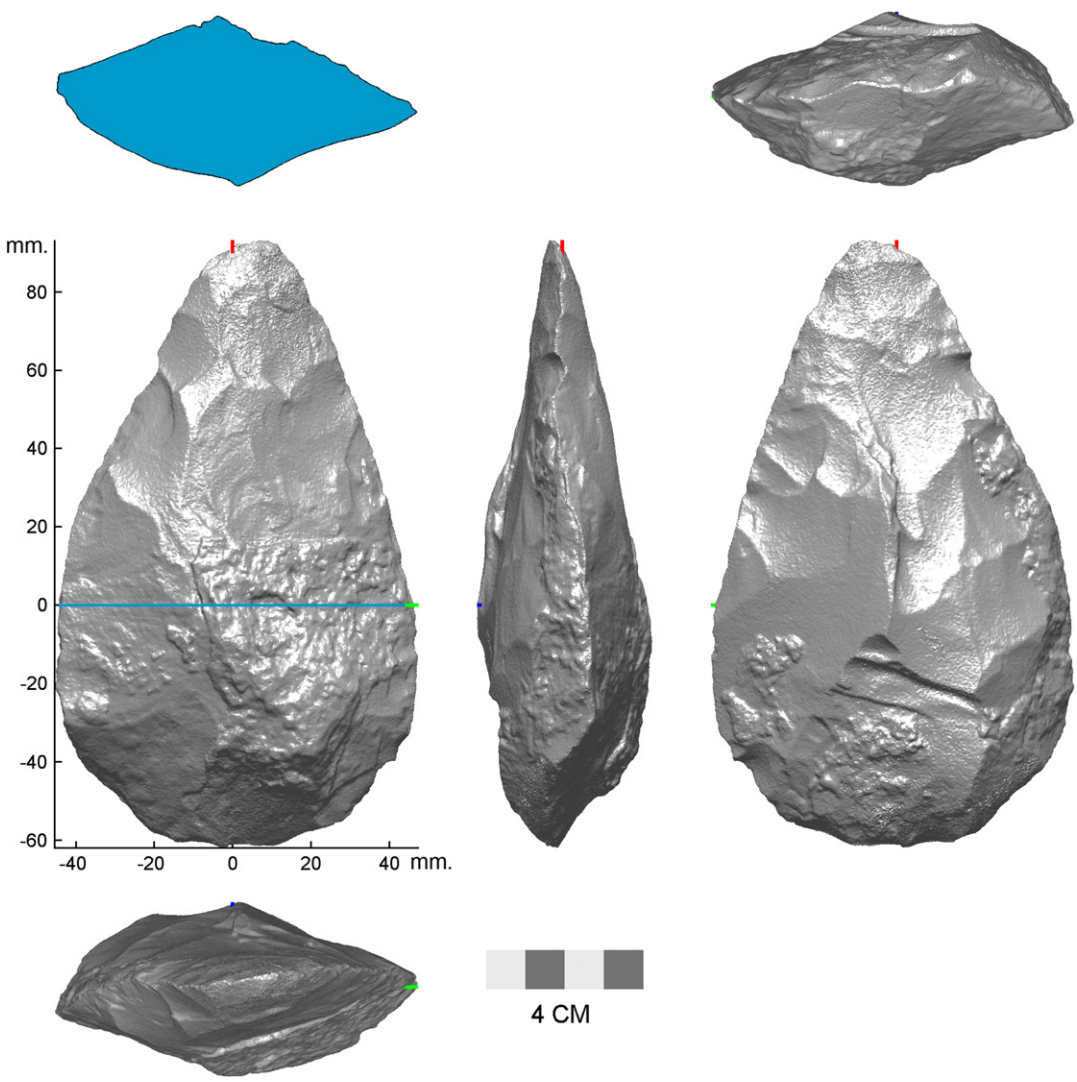


**Fig. 7.** Positioning of a handaxe according to the inertia tensor (including a midsection).

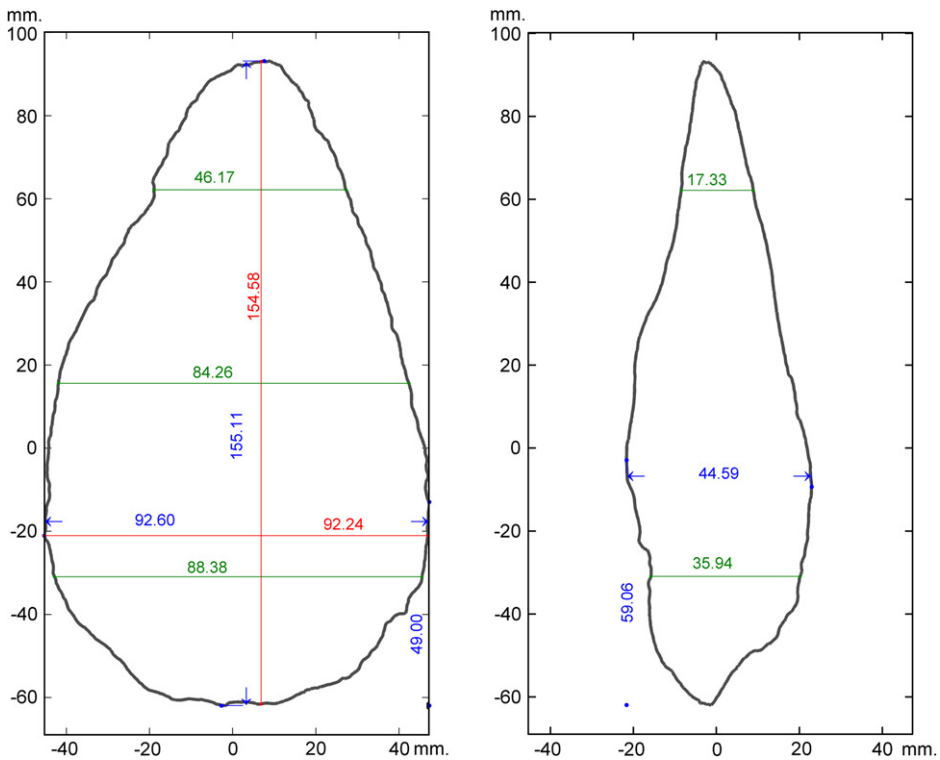


**Fig. 8.** Face and side section of a Handaxe with basic measurements marked.

center of mass (CM) and the coordinates of the center of the enclosing cube (CMEC). The distance between these two points will be denoted by *d*. These parameters provide new data for the typological analysis of these artifacts. We shall use mainly the parameters *v* and *d* in the case studies which follow.

## 5. Case studies

In order to test our method and to illustrate its advantages, a random sample of handaxes was collected from three Lower Paleolithic sites which are spread on a wide geographic area. Included in our sample are handaxes from Ma'ayan Barukh ($N = 39$), Umm Qatafa ($N = 26$), and Nahal Zihor ($N = 25$).

The site of *Ma'ayan Barukh* (MB) is located on the northern slopes descending to the Hula Valley, along the Israel–Lebanon border. The site's assemblage is a surface collection (collected by Amnon Assaf, Kibbutz MB) including ca 6000 handaxes collected from an area of 2 $km^2$. Early analysis (Stekelis and Gilead, 1967) concluded that the entire collection is homogenous and represents a single cultural phase. Gilead (1970a) assigned the MB assemblage to the Late Acheulian, based primarily upon the typology of the handaxes. The availability of a detailed analysis of this assemblage enables us to compare the results of the traditional (manual) and the new method of measurements.

*Umm Qatafa* (UQ) is a karstic cave located in the Judean hills (40 km south of Jerusalem). Neuville's (1951) excavations during the 1930s suggested seven stratigraphic units throughout the archaeological sequence of the cave. The handaxes of UQ were reexamined by Gilead (1970a) who established that the assemblage (layer D2) correlates with the cultural unit of the Late Acheulian.

Apart from a few handaxes drawn by Neuville (1951), there is no detailed database available for contemporary researchers to re-evaluate this important Acheulian industry. The long stratigraphic sequence of UQ is important for anchoring the Late Acheulian to the relative chronological scheme of the Levantine Lower Palaeolithic.

The site of *Nahal Zihor* (NZ) was discovered during recent studies of the arid southern Negev desert which revealed an Early Pleistocene fluvio-lacustrine sequence (Ginat et al., 2003). The paleo-lake sequence indicates that humid climatic conditions during that period supported the formation of a lake (Lake Zihor) which is estimated to have existed ca 100,000 years. Over 100 find-spots and larger occurrences of prehistoric artifacts, primarily handaxes, were discovered in vicinity of the paleo-lake. On the basis of techno-typological and stratigraphic considerations, these assemblages were divided into two groups: the first may be contemporaneous with the lake and is associated with the well known 'Ubeidiya Early Acheulian site' (Nahal Zihor E, NZE), while the second is found mainly on the younger terraces (Ginat et al., 2003) assumed as Late Acheulian occurrences (Nahal Zihor L, NZL) and is associated with the Ma'ayan Barukh site (presented above).

Items of the three collections were scanned and analyzed by the computer-based method presented above. The output is a set of 3-D images and a database which includes the extracted traditional measures and the new attributes which are unique to the 3-D analysis. The UQ and NZ handaxes, which were not reported before, are now documented and available in digital form; an example of the output file of a handaxe from UQ is shown in Fig. 9. The complete database can be found in our group website: http://www.weizmann.ac.il/complex/uzy/archaeomath/handaxe_data.

As previously mentioned, the analysis of the MB collection offers an important methodological step, in which the traditional and the computer-based methods can be compared. This will be reported in Section 6. Following that, we shall make use of the new attributes derived from the 3-D analysis to test the previously proposed temporal assignment of these collections, namely, that the NZE belongs to the Early Acheulian, while the other collections are from the Late Acheulian. This cultural correlation is essential since no absolute dates are available and the dating of these sites is primarily based on the temporal change of handaxes shape (Gilead, 1970a).

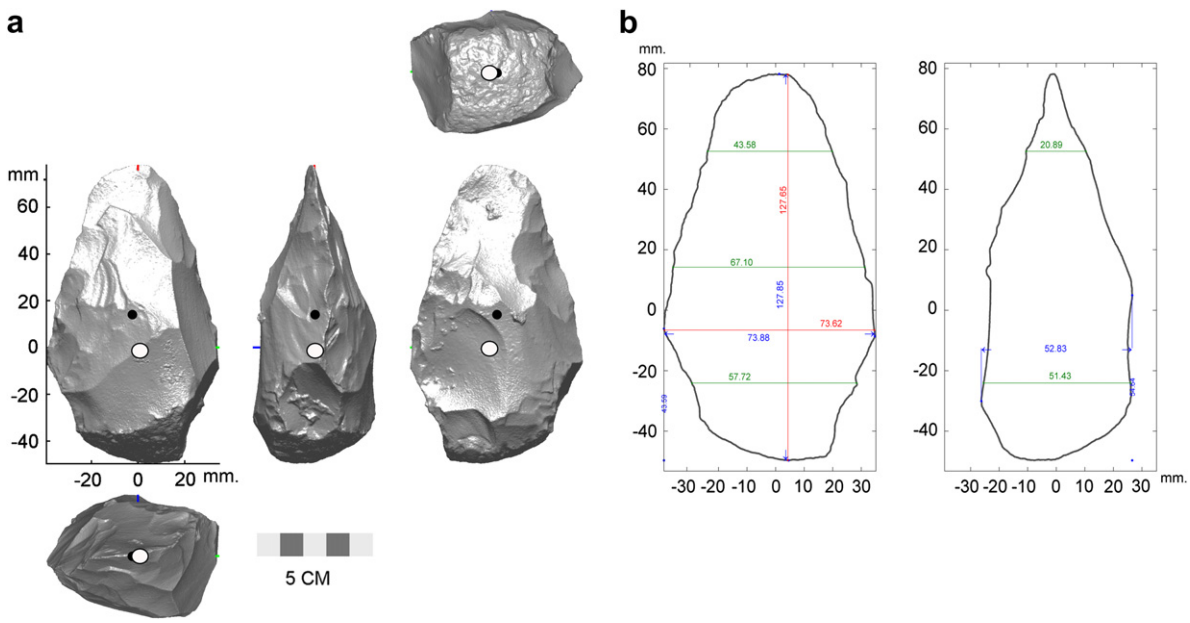


**Fig. 9.** Example of handaxe documentation from Umm Qatafa (#1449). (a) Five views of the handaxe. The location of the center of mass (white dot) and the center of enclosing cube (black dot) are plotted. (b) Basic measurements in a sketch form.

## 6. A comparison of the computerized 3-D analysis and the traditional method

The 'observer' measured twice the same set of handaxes from Ma'ayan Barukh (presented above) which was also scanned and computer analyzed. The comparison between the mean values of the manually extracted measures and the 3-D measures could be carried out on the level of a single handaxe (Fig. 10). In general, the differences between the observer's results and the 3-D analysis were significant. Comparison between length (*L*) parameter shows an average difference of 22.2 mm and a maximum difference of 50.5 mm (Fig. 10a). The differences in one of the handaxes reached 48% of length. Not only that the observer's measurements were inconsistent (Fig. 10b, Table 1), but also differ significantly from the 3-D analysis (Fig. 10b, mean difference 9.3 mm, maximum difference 27.00, minimum difference 0.1 mm).

As expected, larger differences were observed in parameters that are harder to measure manually. For example, the location of the maximum width (L1) requires the observer to measure the maximum width and mark its point on the length axis, while at the same time measure the distance to that point.

Two other traditional databases of handaxes from Ma'ayan Barukh were added for comparison: a collection of 45 handaxes analyzed by Saragusti (2003), and 125 handaxes analyzed by students of the Hebrew University at MB (*HU database*). Although all handaxes are from the same site these databases were measured on handaxes which are not included in the scanned and computer analyzed sample thus cannot be carried out on the level of a single handaxe. However, it is reasonable to assume that the mean values of each variable reflect, in an average way, the deviations between parameters extracted by the various methods. Fig. 11 shows the percentage difference between the three databases and the 3-D analysis in mean values. Once more, the differences are substantial and must be taken into account. Most striking is the difference (28%) between the 3-D scan mean value of L1 (location of maximum width) and the corresponding value in the database of

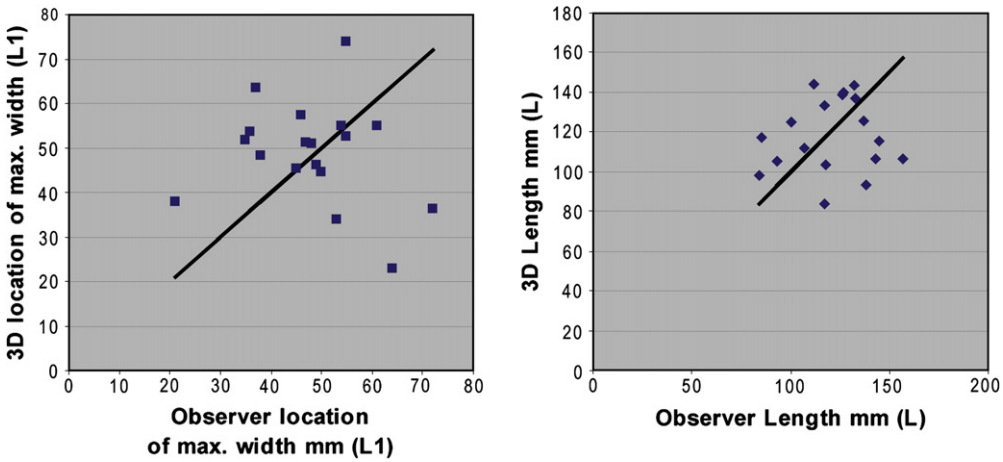


**Fig. 10.** Comparison of the parameters measured by the observer and the 3-D analysis. Each handaxe is represented by a point whose coordinates are the parameters derived by the observer (*x*-axis) and the computerized analysis (*y*-axis). Two parameters are shown: (a) the location of maximum width (L1); (b) the length (*L*). The line represents a complete agreement between the two databases. The farther the points are from the line, the larger is the difference between the measured parameters.

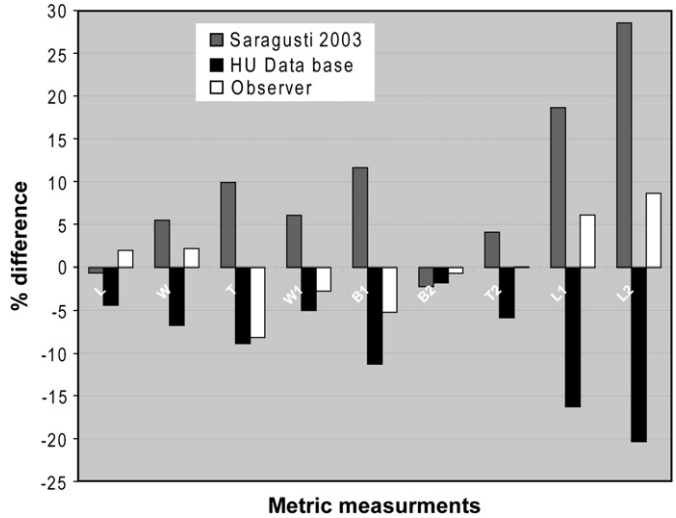


**Fig. 11.** Column bar presenting the difference in mean values between 3-D scans and the databases by attributes. The shorter (minus or plus) the bar the closer the database is to the 3-D values (0).

Saragusti (2003). Even though not all measures differ to the same degree, the variance between the measures illustrates the need for reliable, reproducible, and standardized method for lithic analysis.

## 7. Handaxes typology based on the *v* and *d* parameters – a test of temporal sequencing

The traditional studies of the three sites discussed here suggested a partition of the sites into two distinct temporal groups. The first, NZE, is proposed to be contemporaneous with the well known 'Ubeidiya Early Acheulian site' (Ginat et al., 2003). The second, UQ and NZL are assigned to the Late Acheulian occurrences associated with the MB site presented above (Ginat et al., 2003).

Changes in the dimensions and proportions of the handaxe are frequently correlated with temporal changes. Several metric parameters are chosen in order to measure the dimensional/metrical change. At Misliya Cave, the total length of the handaxe (Zaidner et al., 2006; Gilead, 1970b), at Tabun Cave, the length of the tip as measured from the point of maximum width to the tip (McPherron, 2003), or the length *vs*. the width of the handaxes (Gisis and Ronen, 2006), to quote a few examples.

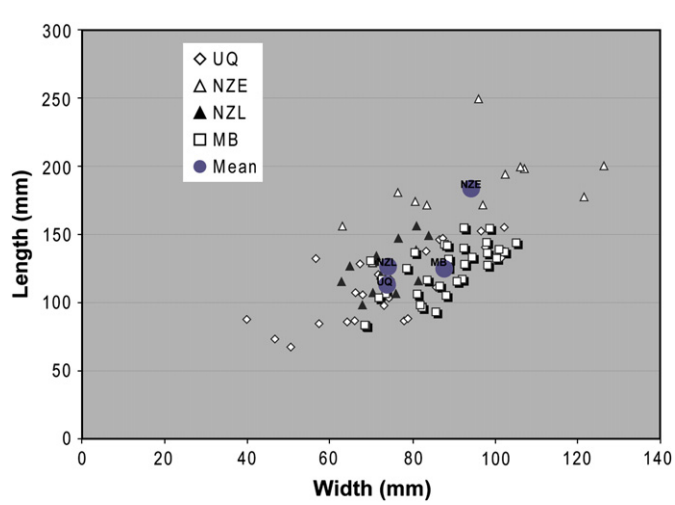


**Fig. 12.** Distribution of the metric parameters *W* (width) and *L* (length) derived from the 3-D analysis of the handaxe assemblages (MB = Ma'ayan Barukh, NZE = Nahal Zihor Early, NZL = Nahal Zihor Late, and UQ = Umm Qatafa).

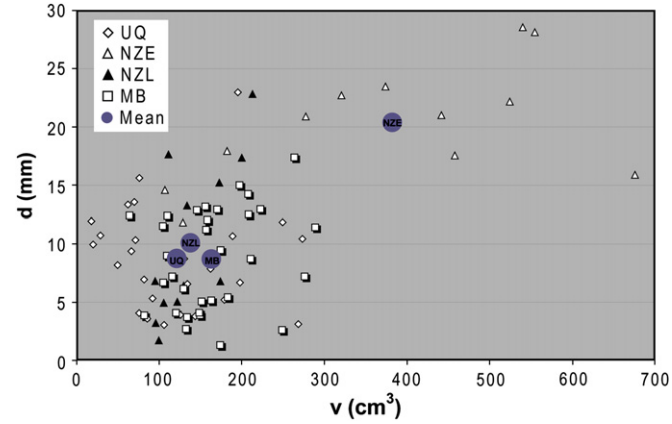


**Fig. 13.** Distribution of the metric parameters *d* (CM–CMEC distance) and *v* (volume) derived from the 3-D analysis of the handaxe assemblages (MB = Ma'ayan Barukh, NZE = Nahal Zihor Early, NZL = Nahal Zihor Late, and UQ = Umm Qatafa).

Here, we address the same problem in two ways. In the first, we use the standard metric parameters derived from the computerized analysis as the basis for the classification, thus following the traditional method of analysis. In the second, we use the volume *v*, and the distance between the CM and the center of the enclosing cube, *d*, to characterize the artifacts. These parameters can be extracted only by the method discussed here.

In Fig. 12 each of the handaxes in the scanned assemblage is represented by a point in the (width, length) plane. The handaxes of each collection are identified by different symbols and their mean positions are also marked. Although the mean value of NZE stands apart from other values, in particular with respect to the length parameter, still there is no clear-cut difference between the NZE and other collections based on this attribute.

In the second stage, we performed a similar analysis, now based on the parameters v and *d* which are unique to the 3-D analysis. Fig. 13 summarizes the results of this analysis. In general, the location of the center of mass in all handaxes is close to the center of the enclosing box (maximum distance of 30 mm). However, the volumes of the NZE handaxes differ substantially from the other collections. This analysis enhances the results obtained by the analysis of the *W* and *L* parameters. It demonstrates a general time-related trend for the tested attributes and supports the relative chronology previously suggested: The MB, UQ and NZL seem to cluster within the same typological (and hence temporal) group, while the early NZE belongs to a different typological group. As for the interpretation we strengthen the previously suggested interpretations.

It is interesting to note that the handaxes from all the Late Acheulian groups, which appear in different sizes, have different shapes, vary in the degree of refinement and are manufactured using a variety of raw materials and technologies. Yet, they all cluster about a common small domain in the (*v*, *d*) plane.

## 8. Summary

This study presented a new method for the documentation and the extraction of metric parameters for lithic artifacts. We have demonstrated its unique features and its potential advantages by applying it to an assemblage of 90 handaxes. The test case has illustrated the process of scanning, recording and the subsequent analysis which consists of several steps: establishing an algorithmically defined positioning, producing 2D drawings of any desired view or section, and extracting metric measurements, traditional as well as those which can be derived only when the complete 3-D description of the object is available.

Archaeologists wish to quantify shapes of artifacts and test their similarity by a set of parameters. We showed the traditional linear measures, which are manually extracted, are biased by the ambiguities inherent to positioning, as well as by human errors. The 3-D methodology presented above offers an automatic and bias-free method for extracting the same parameters. Moreover, once the digital model of the artifact is available, other parameters can be extracted. In the test cases described here, the clustering to distinct groups is sharper when these parameters are used as the basis for the classification.

The main practical drawback for implementing this technology is the high cost of the equipment. The initial investment is paid off by both the reduced costs for obtaining technical drawings of artifacts, and by the additional information gained. It is clear that the use of the 3-D scanner becomes economically viable when it is used intensively by a several groups of archaeologists.

Finally, the advantage of this methodology is that it results with a digital database which includes the 3-D images (point-cloud), their projections and sections, and the metric parameters extracted from the analysis. This database can be easily distributed so that the community of researchers will be able to share complete databases allowing an easy and better exchange of information.

Our digital data are available on the website of the Computerized Archaeology group at Weizmann Institute of Science: http://www.weizmann.ac.il/complex/uzy/archaeomath/handaxe_data.html.

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